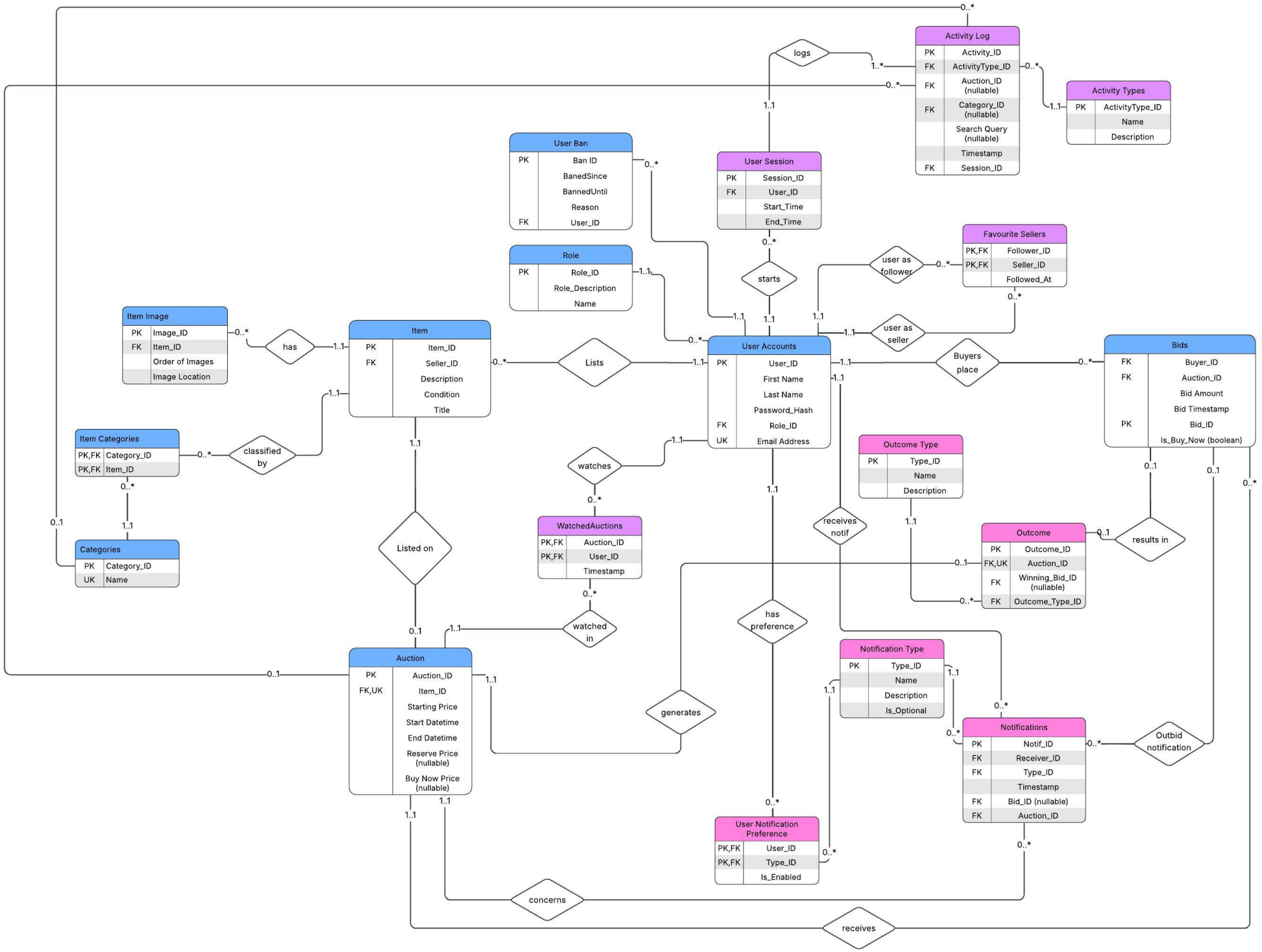

Activity Log
PK Activity_ID
FK ActivityType_ID
FK Auction_ID (nullable)
FK Category_ID (nullable)
Search Query (nullable)
Timestamp
FK Session_ID
Activity Types
PK ActivityType_ID
Name
Description
logs
User Ban
PK Ban ID
BanedSince
BannedUntil
Reason
FK User_ID
User Session
PK Session_ID
FK User_ID
Start_Time
End_Time
starts
Role
PK Role_ID
Role_Description
Name
Favourite Sellers
PK,FK Follower_ID
PK,FK Seller_ID
Followed_At
user as follower
user as seller
Item Image
PK Image_ID
FK Item_ID
Order of Images
Image Location
has
Item
PK Item_ID
FK Seller_ID
Description
Condition
Title
Lists
User Accounts
PK User_ID
First Name
Last Name
Password_Hash
FK Role_ID
UK Email Address
Buyers place
Bids
FK Buyer_ID
FK Auction_ID
Bid Amount
Bid Timestamp
PK Bid_ID
Is_Buy_Now (boolean)
Item Categories
PK,FK Category_ID
PK,FK Item_ID
classified by
Categories
PK Category_ID
UK Name
watches
WatchedAuctions
PK,FK Auction_ID
PK,FK User_ID
Timestamp
Listed on
Outcome Type
PK Type_ID
Name
Description
receives notif
Outcome
PK Outcome_ID
FK,UK Auction_ID
FK Winning_Bid_ID (nullable)
FK Outcome_Type_ID
results in
watched in
has preference
Auction
PK Auction_ID
FK,UK Item_ID
Starting Price
Start Datetime
End Datetime
Reserve Price (nullable)
Buy Now Price (nullable)
generates
Notification Type
PK Type_ID
Name
Description
Is_Optional
Notifications
PK Notif_ID
FK Receiver_ID
FK Type_ID
Timestamp
FK Bid_ID (nullable)
FK Auction_ID
Outbid notification
User Notification Preference
PK,FK User_ID
PK,FK Type_ID
Is_Enabled
concerns
receives

**Explanation of our ER Diagram:**

**User Notification Preferences**

Users have the ability to state their preference on what kind of notifications they would like to get (e.g. users may choose to not receive notifications when their favourite seller posts a new auction or changes in status of the items on their watchlist).

**Enums**

**Notification Types:**

- **For Buyer:**
    - Outbid by another bidder
    - Followed seller has a new item on auction
    - New auction based on preferences
    - Auction start/end reminder
    - Outcome of auction
    - Successful completion of Buy Now Bid
- **For Seller:**
    - New bid on item on auction
    - New follower
    - Item is on watchlist
    - Outcome of auction
    - Complaint registered by system
    - Feedback registered by system

**Outcome Types:**

- Item sold to highest bidder
- Item sold to “buy now” bidder
- Unsold
- Cancelled auction
- Admin shut down (this is dependent on whether we can have an admin role)

**Roles:**

- Buyer
- Seller
- Admin

**Filters for Items:**

- Categories
- Price Range
- Item Name Search Query
- Condition (second hand/used, like new, new, heavily used, etc.)

- Whether listings have images
- Status of Auction (Recently started, Ending soon, Brand New, Closed, Active, etc.)